2020秋高数上期末(回忆版)

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) The number of the real roots for the equation $x^3-3x+3=0$ is

(A) 0 (B) 1 (C) 2 (D) 3

(2) If $f(x)$ is continuous on $(-\infty,+\infty)$, which of the following statements is wrong?

(A) $\int_0^x f(x)\,dx=\int_0^t f(t)\,dt$.

(B) $\int_0^t f(x)dx=\int_0^t f(\sin x)\,d(\sin x)$.

(C) $d\left(\int_0^x f(t)\,dt\right)=f(x)\,dx$.

(D) $d\left(\int_0^{x^2} f(t)\,dt\right)=f(x^2)\,d(x^2)$.

(3) Let $f(x)=\begin{cases}x^4\sin\frac{1}{x}, & x\neq 0\\ 0, & x=0\end{cases}$

Then the largest positive integer $n$, for which $f^{(n)}(0)$ exists, is

(A) 1 (B) 2 (C) 3 (D) 4

(4) If $f(x)$ is twice-differentiable on $(-\infty,+\infty)$, and $g(x)=(1-x)f(0)+xf(1)$, then which of the following statements is correct on $(0,1)$?

(A) $f(x)>g(x)$ if $f'(x)>0$.

(B) $f(x)>g(x)$ if $f'(x)<0$.

(C) $f(x)>g(x)$ if $f''(x)>0$.

(D) $f(x)>g(x)$ if $f''(x)<0$.

(5) If the improper integal $\int_0^{+\infty}\frac{\tan^{-1}(x^2)}{x^k}\,dx$ converges, then the constant $k$ must satisfy

(A) $k<1$ (B) $k>3$ (C) $1<k<2$ (D) $1<k<3$.

2. Fill in the blanks.

(1) function $f(x)=x^2$ has a tangent line $y=kx-1$ if $K=$_____, or _____.

(2) Assume that $f'(0)=3$, $f''(0)=5$, $f'(1)=-4$, and $f''(1)=-7$. Let $g(x)=f(\ln x)$. Then $g''(1)=$_____.

(3) The avergge value for $f(x)=\sin^2 x$ on $[0,\pi]$ is _____.

(4) Let $y=(\cos x)^x$ for $0<x<\frac{\pi}{2}$, then $y'(x)=$_____.

(5) If $f''(a)$ exists, and $f'(a)\neq 0$, then $\lim_{x\to a}\left(\frac{1}{f'(a)(x-a)}-\frac{1}{f(x)-f(a)}\right)=$_____.

3. The region D is enclosed by the curve $y=\ln\sqrt{x-1}$, the straight line $x=5$, and the x-axis.

(1) Find the area of the region D.

(2) Find the volumes generated by revolving the region D about the line $x=5$.

4. Find the particular solution of

$$xy'+(x-2)y=3x^3e^{-x},\quad x>0,$$

satisfying $y(1)=0$.

5. Evaluate the following limits.

(1) $\lim_{n\to+\infty}\left(\frac{n}{2n^2+3n+1^2}+\frac{n}{2n^2+6n+2^2}+\cdots\frac{n}{2n^2+3nk+k^2}+\cdots+\frac{n}{2n^2+3n^2+n^2}\right)$

(2) $\lim_{x\to 0}\left(\frac{\ln(1+x)}{x}\right)^{\frac{1}{e^x-1}}$.

6. (1) For $y=\frac{x^2+1}{x+1}$, identify the coordinates of any local and absolute extreme points and inflection points that may exist.

(2) Sketch the graph of the above function. (Please identify all the asymptotes and some specific points, such as local maximum and minimum points, inflection points, and intercepts.)

7. Find $\frac{dy}{dx}$ if $y=\int_{x^2+1}^{2x^2+3} t\tan\sqrt{x+t}\,dt$.

8. Evaluate the integrals.

(1) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \csc^3 x\,dx$.

(2) $\int \sqrt{\frac{x}{x-2}}\,dx$, where $x>2$.

(3) $\int_1^e \ln^3 x\,dx$.

(4) $\int_1^{+\infty} \frac{(x+2)\ln(x^2+1)}{x^3}\,dx$.

9. Let $f(n)=\sum_{m=1}^{n}\int_0^m \cos\frac{2\pi n\lfloor x+1\rfloor}{m}\,dx$, here $\lfloor x+1\rfloor$ is the largest integer which is less than or equal to $x+1$. Evaluate $f(2021)$.